



Cosmology

Galaxies and Beyond

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Overview

What is Cosmology?

Our Place in the Cosmos

The Big Bang

Dark Matter

Dark Energy

$t = 0$

Modern Astronomy

What is Cosmology?

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Shape: What happens as you travel across the universe?

Content: What is Dark Matter and Dark Energy?

The Night Sky

The Milky Way



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The Milky Way



Ground based telescopes

The Coma Cluster



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The Coma Cluster



Space Telescopes

Hubble Extreme Deep Field



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The Dark Ages

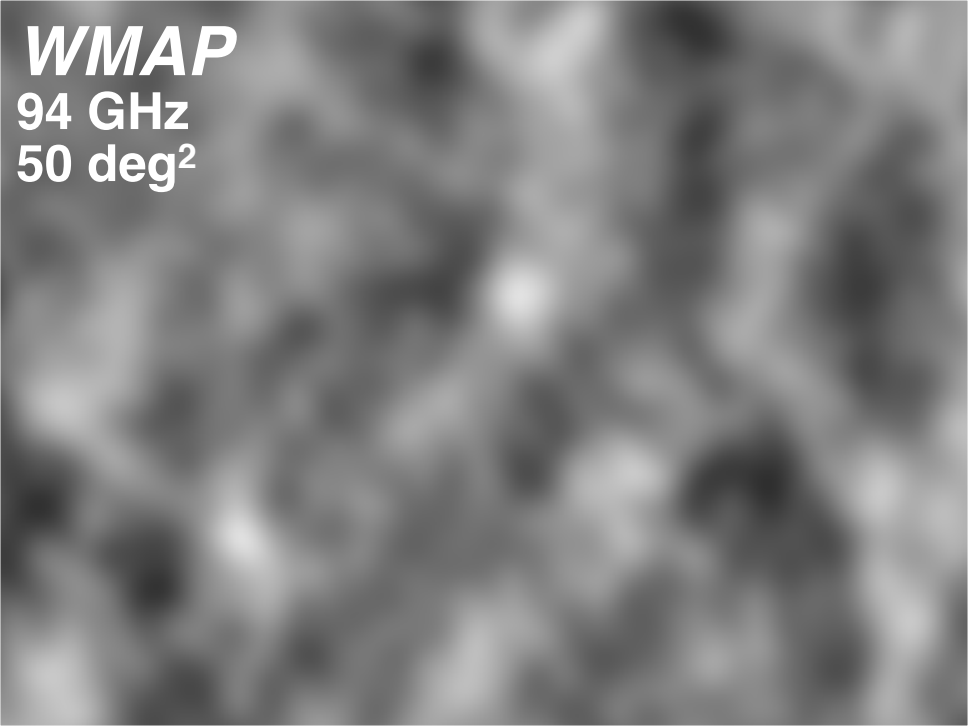
The CMB

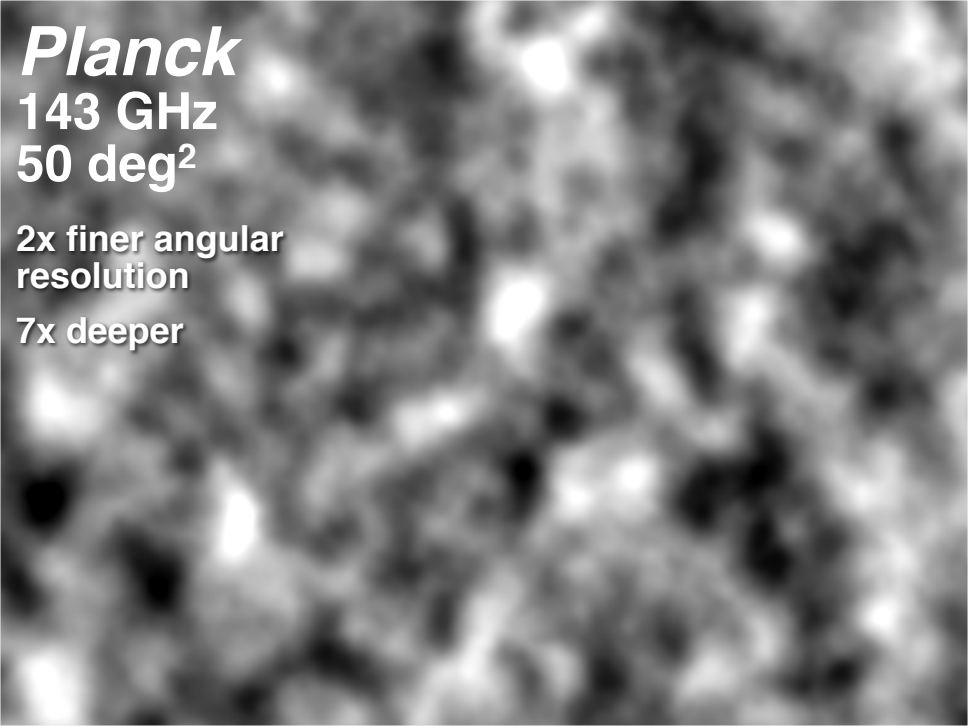
Cosmic Microwave Background

WMAP

94 GHz

50 deg²





Planck

143 GHz

50 deg²

**2x finer angular
resolution**

7x deeper

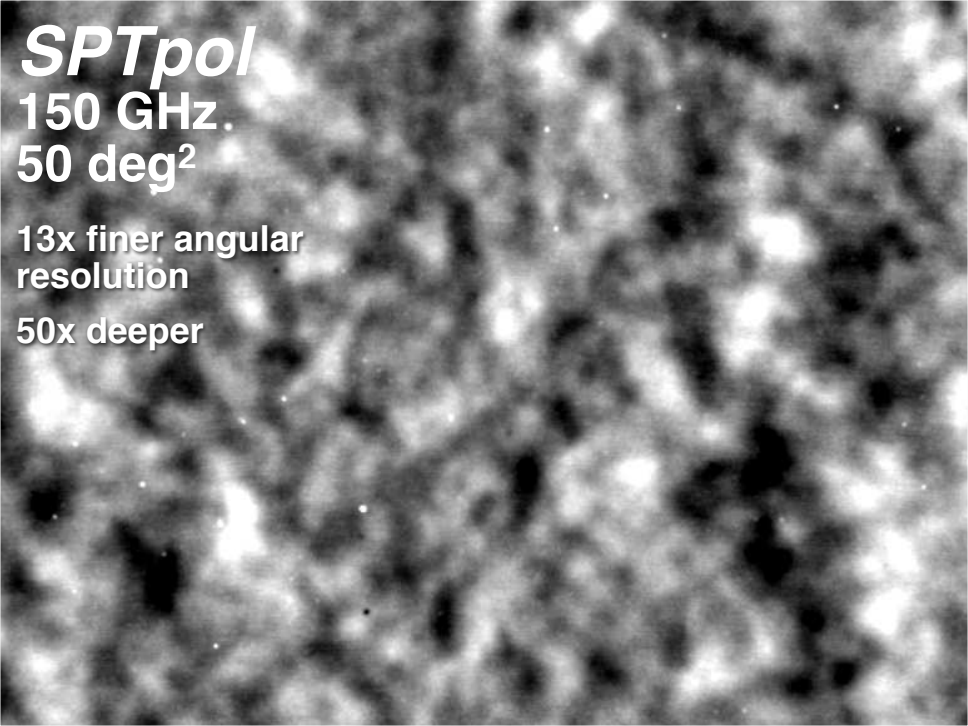
SPTpol

150 GHz.

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What is the CMB?

- ▶ In order to understand what this CMB radiation is, we need to dispel some common misconceptions associated with “The Big Bang”.

The Big Bang

Galactic Redshift



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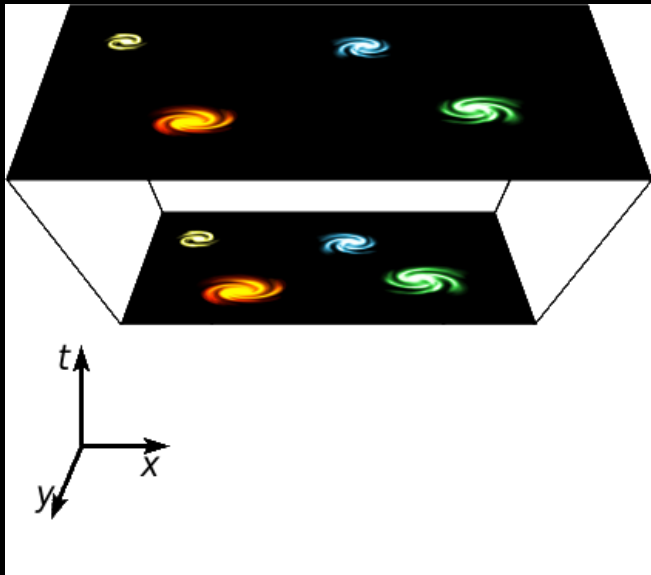
Galactic Redshift

As observed by Edwin Hubble in 1929, it appears that:

1. All Galaxies are moving away from us
2. The Further away they are, the faster they are moving away
3. If we were to go to another galaxy, we would see exactly the same thing

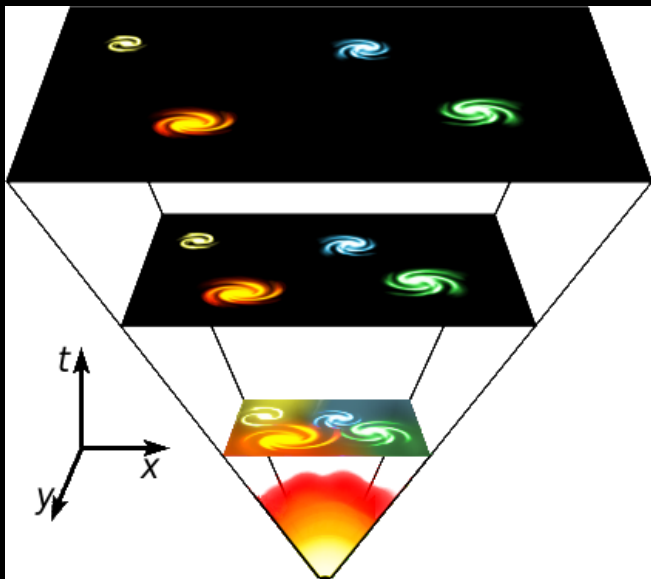
The Big Bang

A stretching universe:



The Big Bang

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The Big Bang

Common Misconceptions

The Big Bang does *NOT* look like this:



The Big Bang

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- ▶ The big bang didn't begin at a particular point

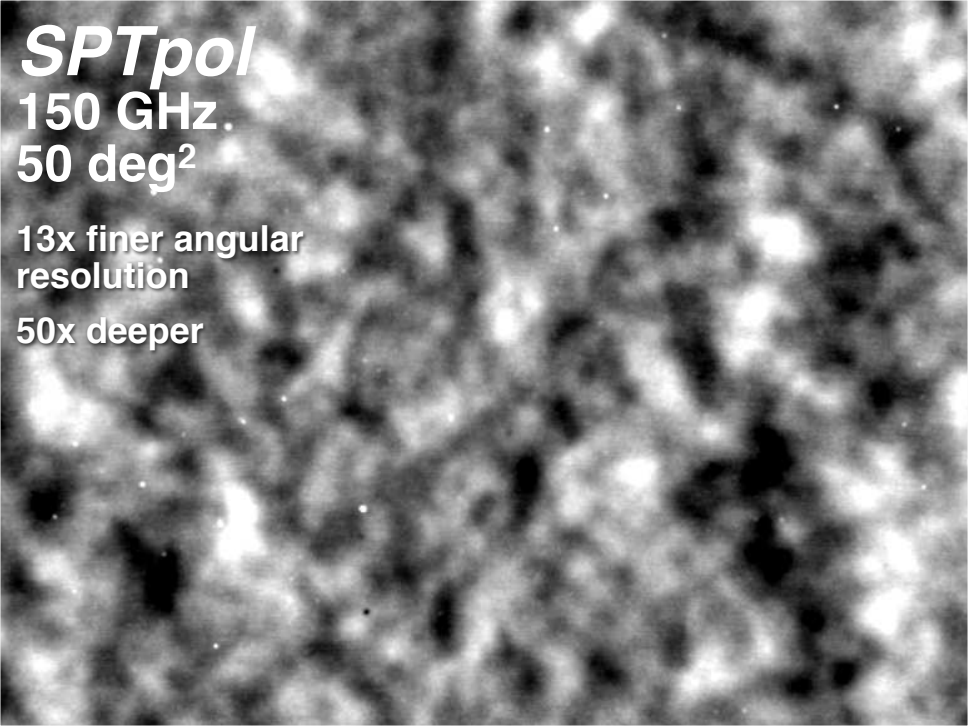
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Current Issues in Cosmology

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Dark Matter

The Problem



Dark Matter

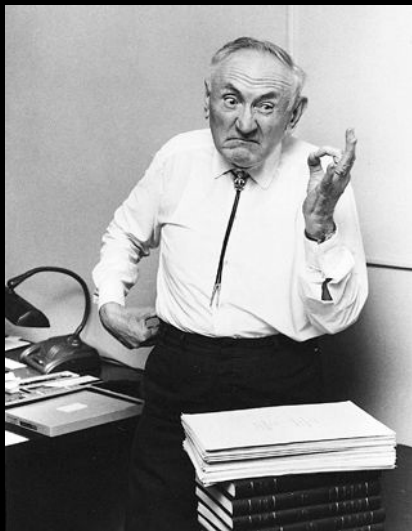
The Problem



Dark Matter

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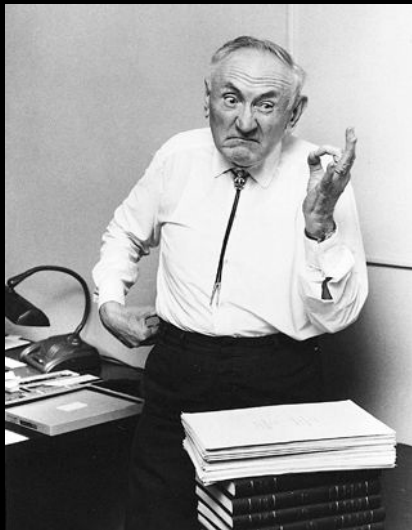
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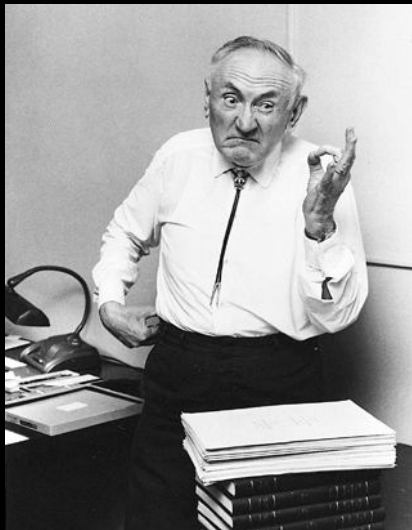
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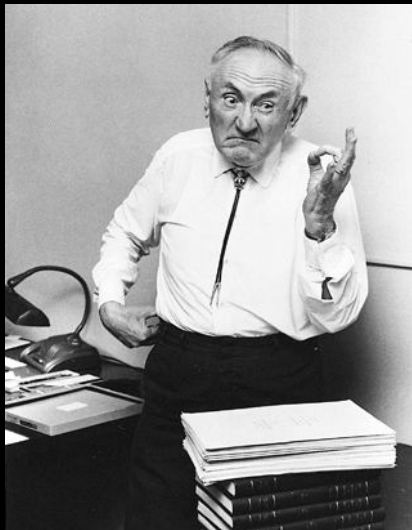
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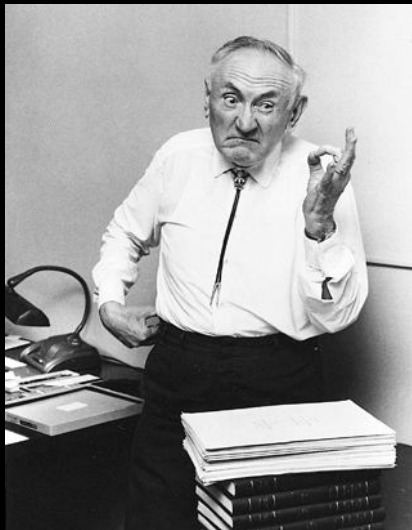
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- ▶ There are many other reasons for believing in Dark Matter.
- ▶ Much effort has been put into trying to find the “missing matter” -but so far to no avail.
- ▶ Right now it's really just a name for a phenomenon we don't understand yet.



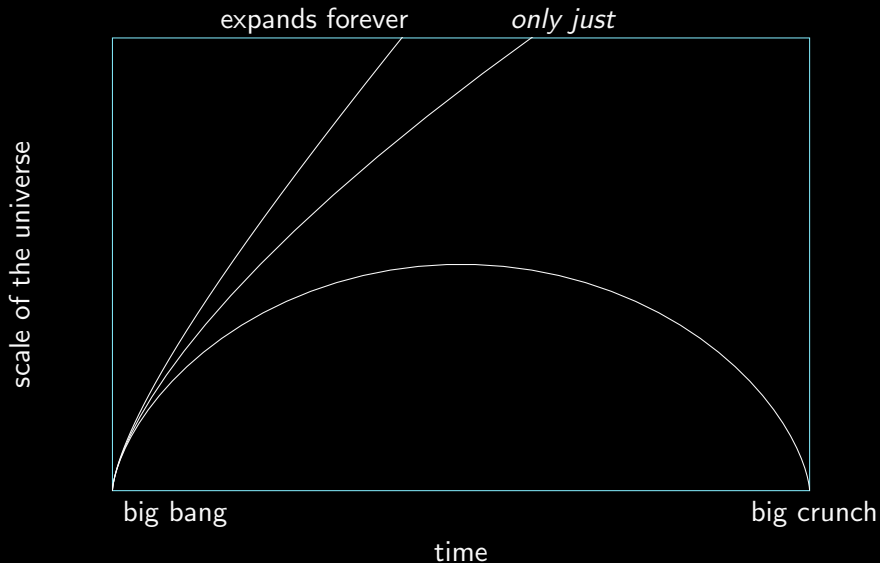
Dark Energy

The Problem



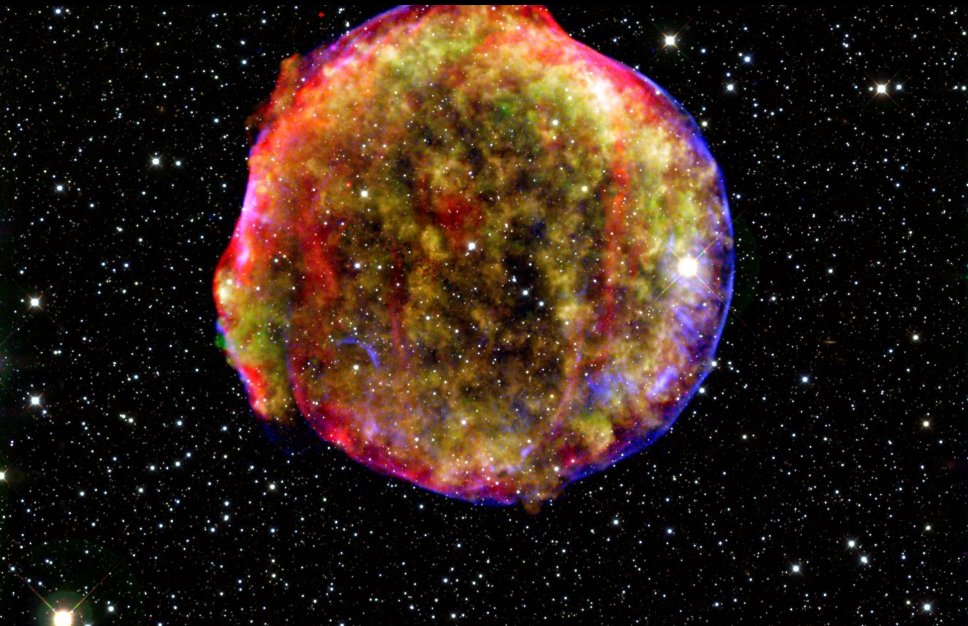
The evolution of the universe

Three options:



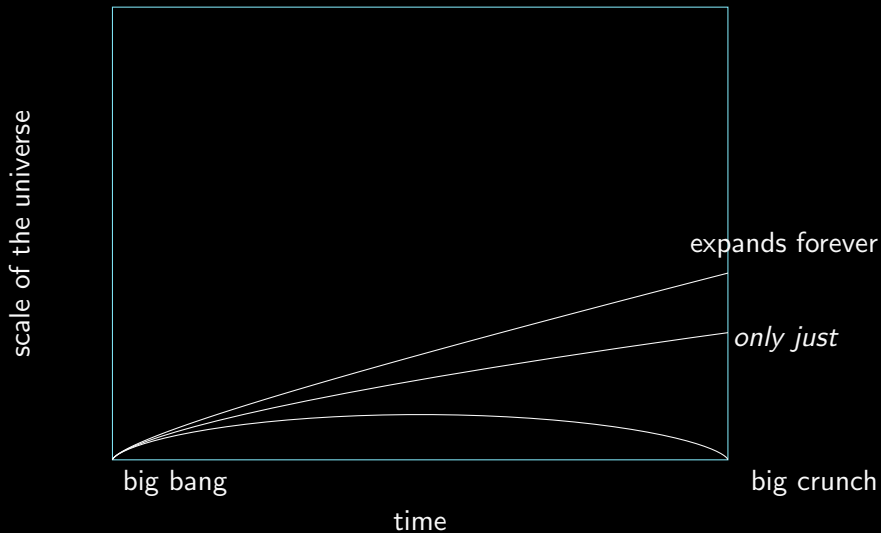
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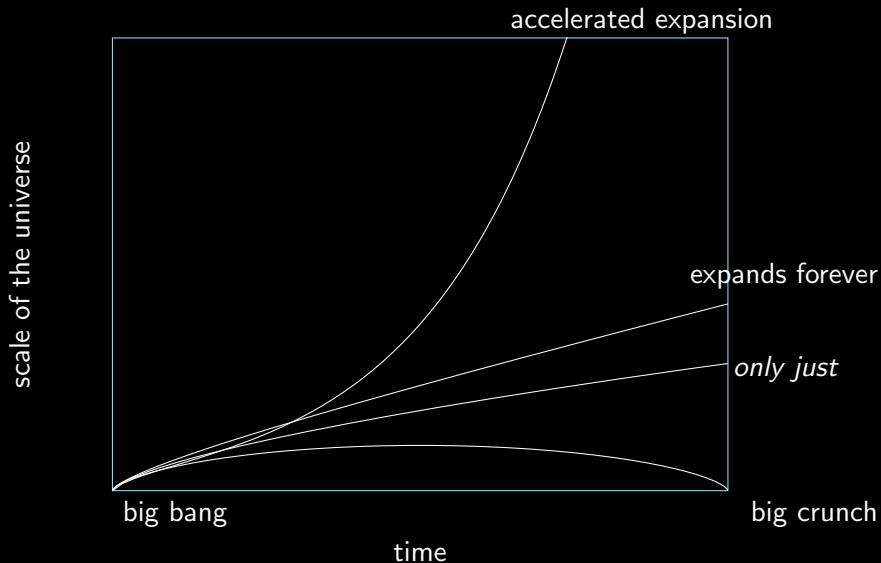
The evolution of the universe

Three options:



The evolution of the universe

None of the above:



Dark Energy

The Problem

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Dark Energy

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Dark Energy

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- ▶ One of the ways in which this could happen would be the presence of an energy permeating the universe, accelerating the expansion.
- ▶ We call this Dark Energy
- ▶ Like Dark Matter, it's really just a name for a phenomenon which we observe, and haven't been able to explain yet

Dark Energy...

...or Cosmological Constant?

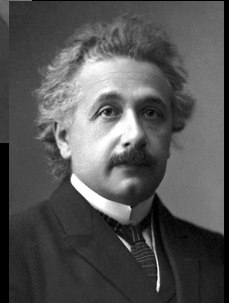
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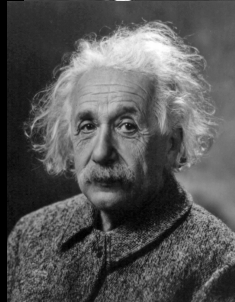
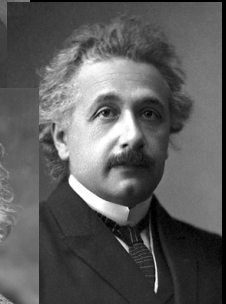
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Dark Energy...

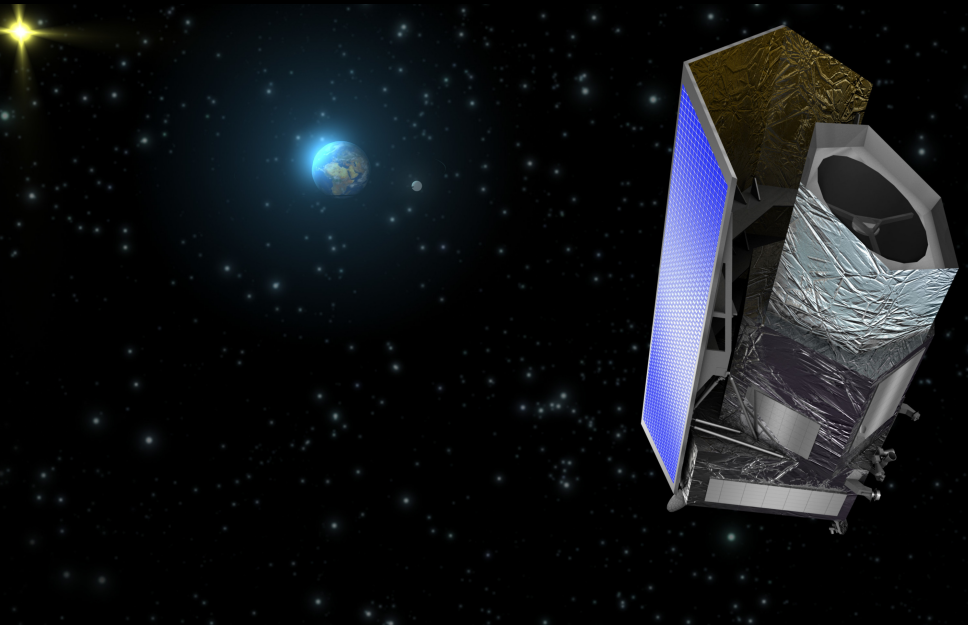
...or Cosmological Constant?

- ▶ Einstein's gravity can be repulsive at large scales.
- ▶ This is known as a cosmological constant.
- ▶ In modern terms, we think of it as a type of energy.



The Euclid Satellite

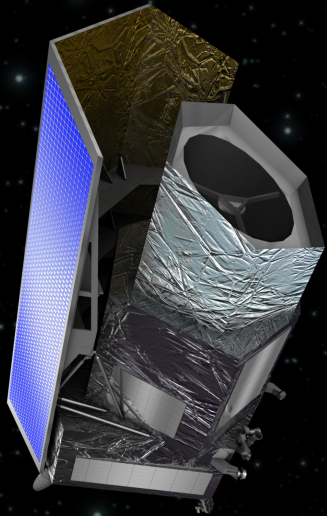
Exploring dark energy



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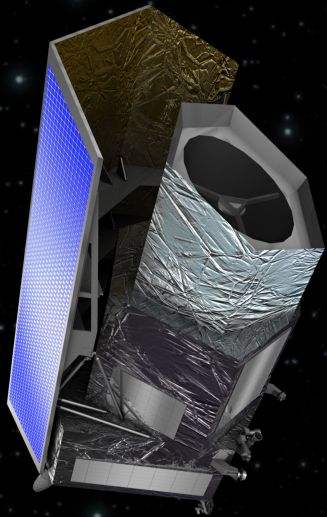
► Launch date $\approx$ 2020



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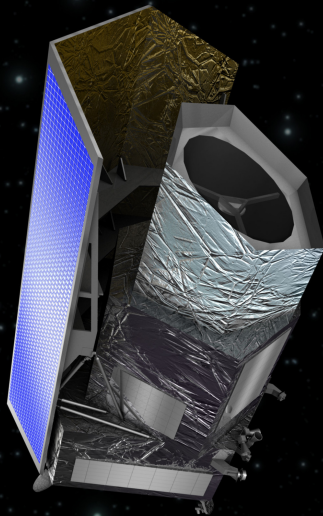
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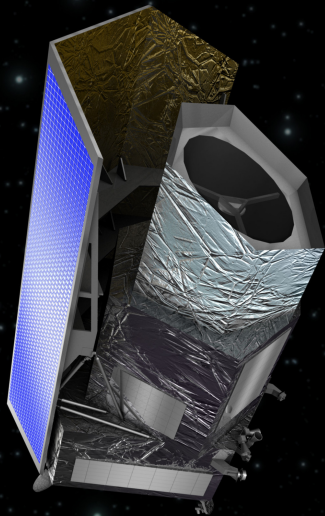
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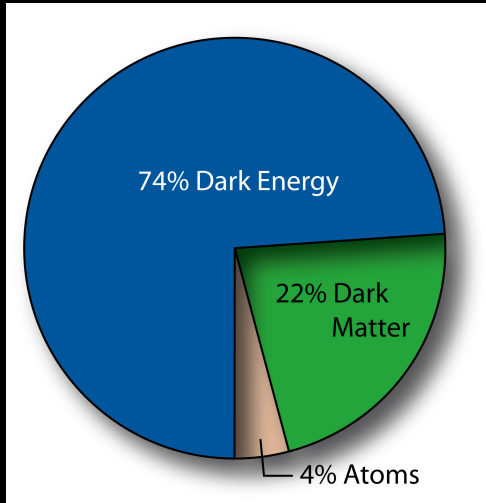
The Euclid Satellite

Exploring dark energy

- ▶ Launch date $\approx$ 2020
- ▶ Build up an accurate 4D picture of the visible universe
- ▶ By measuring cosmic shear around galaxies and clusters, will be able to build up a map of Dark Matter
- ▶ Will see any variation in dark energy across space and time, enabling us to build a better understanding of it.

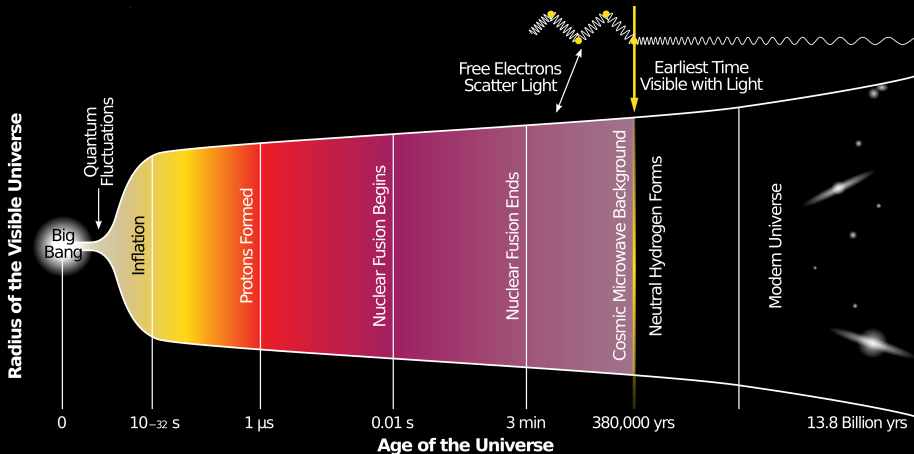


The Composition of the Universe



The beginning of time

$t = 0$



Inflation

A convenient explanation

1. The CMB is far too uniform

Inflation

A convenient explanation

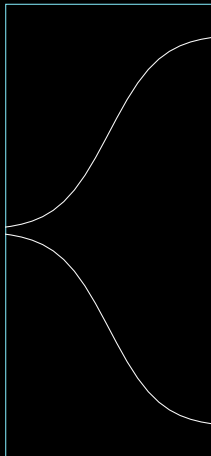
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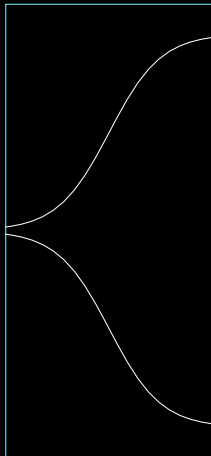
time

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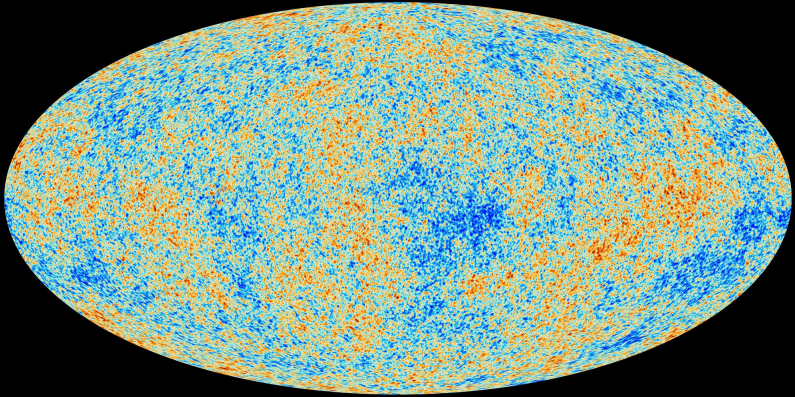
1. The CMB is far too uniform
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3. We call this accelerated expansion “Inflation”

scale of the universe



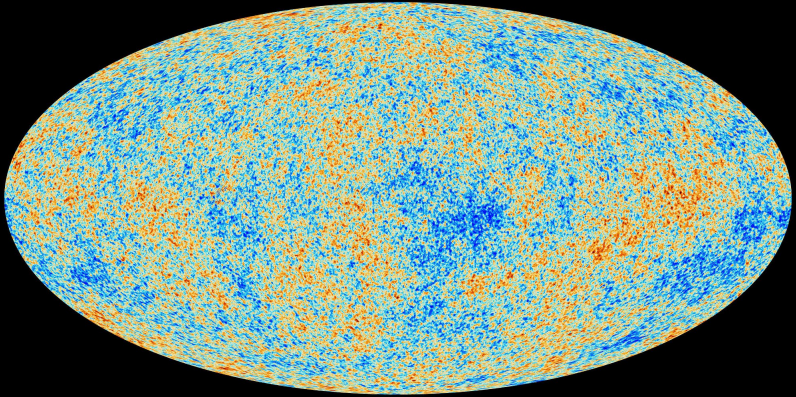
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Anisotropies in the CMB



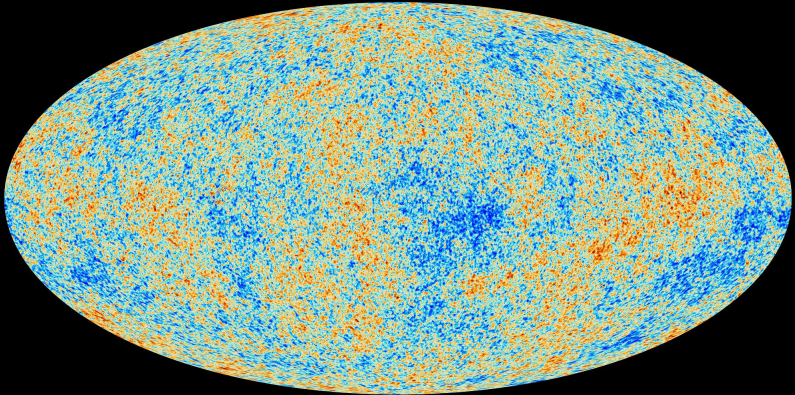
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- ▶ The *Theory of Inflation* predicts these inhomogeneities,
- ▶ Which are created by *quantum fluctuations*.
- ▶ These are the seeds of structure formation in the universe.

The dawn of a new era in astronomy

The rise of the mega-telescope

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The rise of the mega-telescope

- ▶ Square Kilometre Array

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- ▶ Planck Satellite

Square Kilometre Array (SKA)

2019: World-wide radio telescope



SPDO / Square Kilometre Array Project

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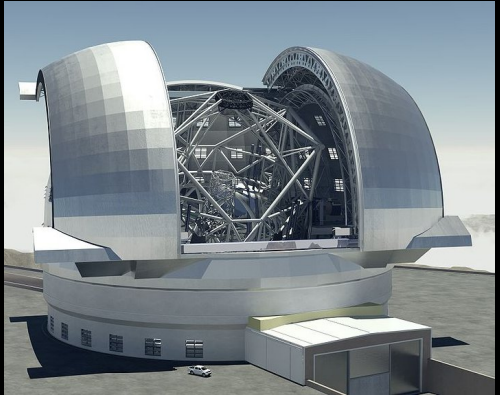
2019: World-wide radio telescope



- ▶ Cost: €1.5bn
- ▶ Testing GR, DE, DM, Magnetism...
- ▶ Imaging Hydrogen through the universe will build a 3D picture of the first ripples of structure.

European Extremely Large Telescope (E-ELT)

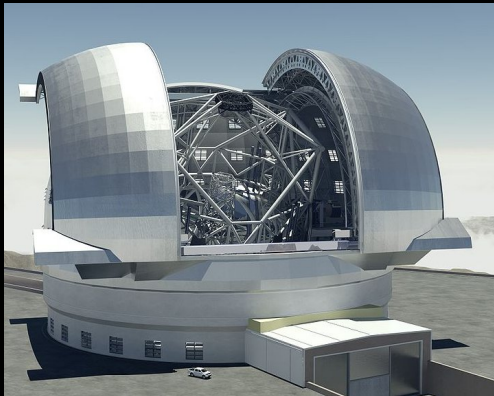
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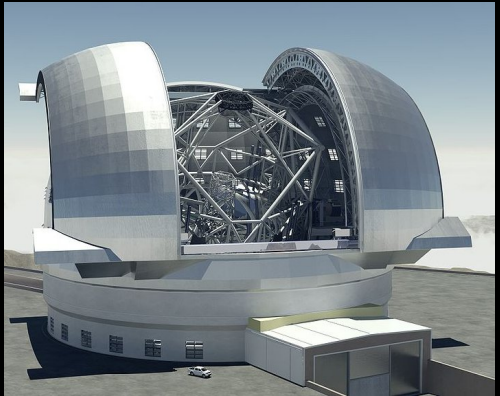
► Cost $\approx$ €1.3bn



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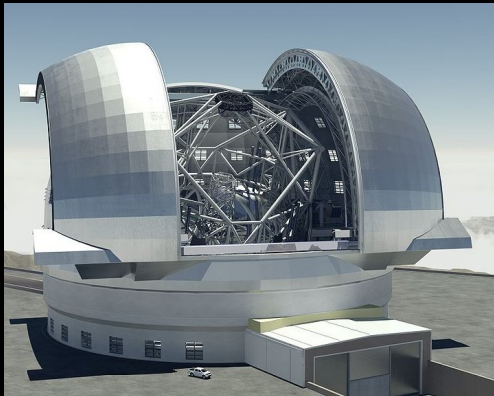
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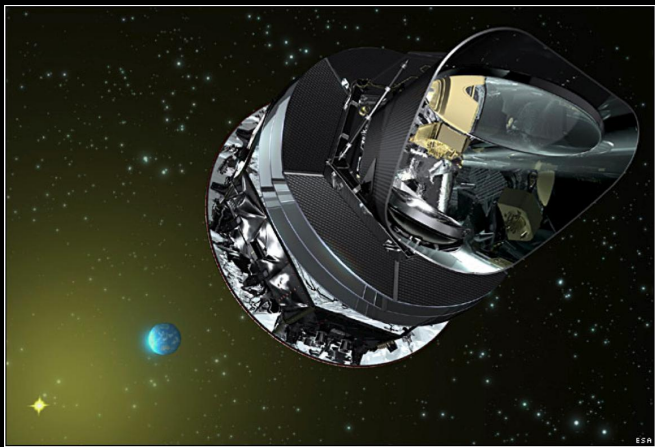
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- ▶ Over a 20 year baseline, it will be sufficiently accurate to watch the universe expand.



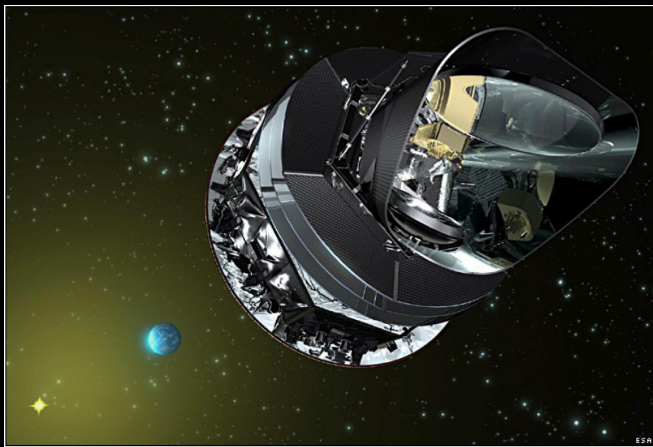
The Planck Satellite

In operation: Space-based microwave telescope



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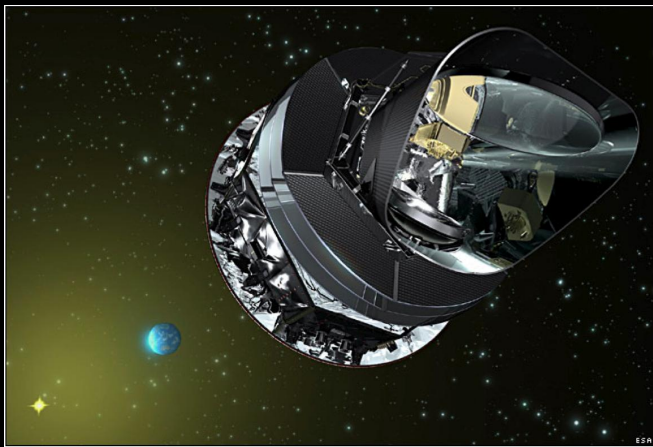
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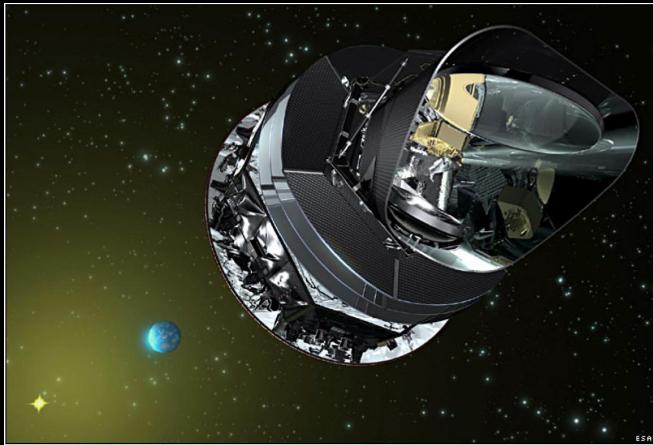
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- ▶ Next-generation CMB mapping: polarisation
- ▶ Cataloguing galaxy clusters

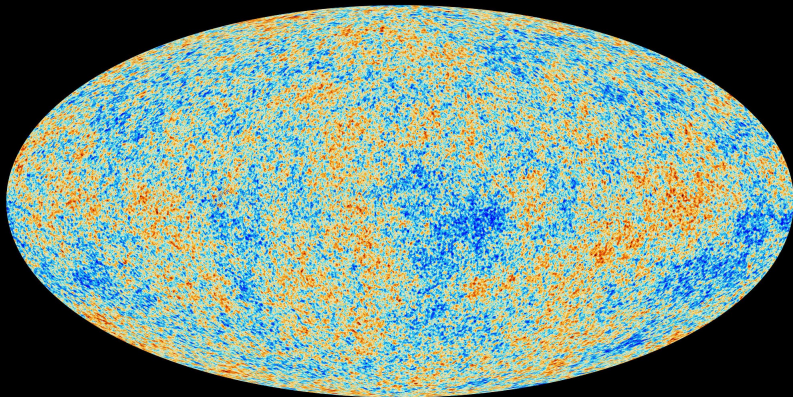
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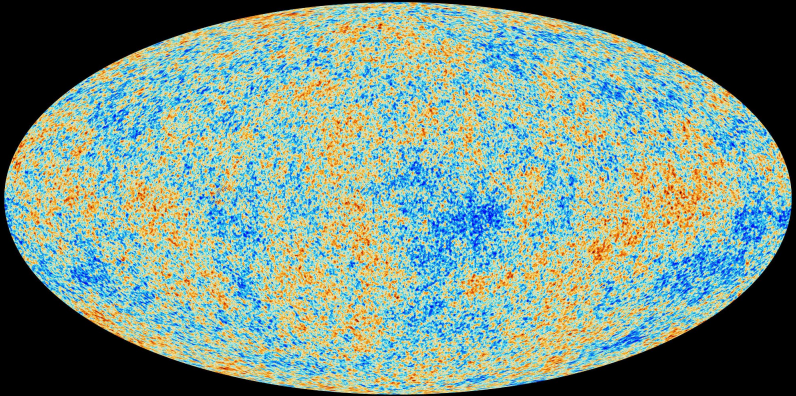


- ▶ Next-generation CMB mapping: polarisation
- ▶ Cataloguing galaxy clusters
- ▶ Providing essential data for accurate exploration of cosmological theories

Summary

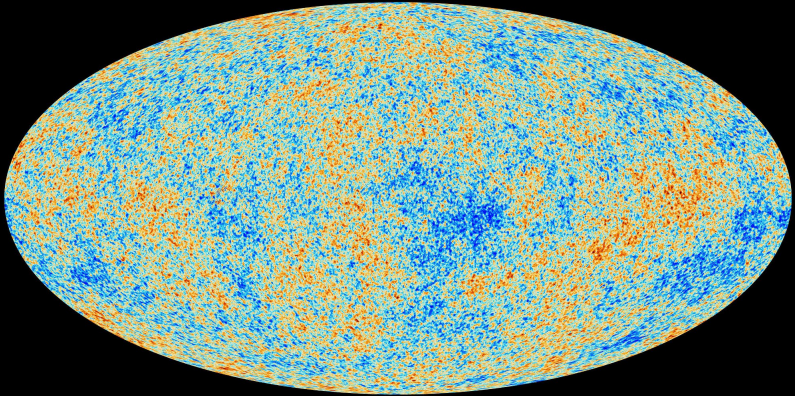


Summary



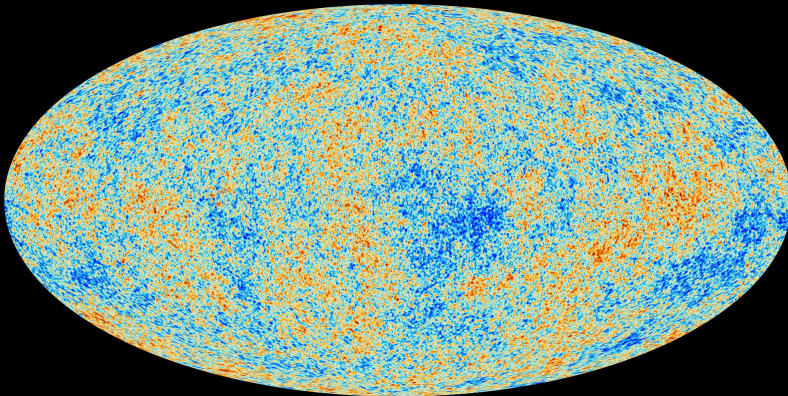
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Summary



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- ▶ There are many unsolved theoretical problems,
- ▶ And a wealth of data to explore them with.

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Age: How old is the universe?

Birth: Is there a beginning of time? What was it like?

Fate: Will everything come to an end?

Size: How big is the cosmos? Is there an edge?

Shape: What happens as you travel across the universe?

Content: What is Dark Matter and Dark Energy?